

TAREA 1.

ALGEBRA DE MATRICES UTILIZANDO R

Problema 1.

Sea $A = \begin{pmatrix} 7 & -3 & 2 \\ 4 & 9 & 5 \end{pmatrix}$

- Encuentre A^t
- Encuentre $(A^t)^t = A$
- Obtenga $A^t A = A A^t$

Rcode

```
#-----
> #Problema 1
> rm(list=ls())
> #Sea
> (A<-matrix(c(7,4,-3,9,2,5),nrow = 2,ncol = 3))
      [,1] [,2] [,3]
[1,]    7   -3    2
[2,]    4    9    5

> # Encuentre A'
> (At<-t(A))
      [,1] [,2]
[1,]    7    4
[2,]   -3    9
[3,]    2    5

> # Encuentre (A')'=A
> t(At)
      [,1] [,2] [,3]
[1,]    7   -3    2
[2,]    4    9    5
> (t(At)==A)
      [,1] [,2] [,3]
[1,] TRUE TRUE TRUE
[2,] TRUE TRUE TRUE

> # Obtenga A'A=AA'
> At%%A
      [,1] [,2] [,3]
[1,]   65   15   34
[2,]   15   90   39
[3,]   34   39   29
> A%%At
      [,1] [,2]
[1,]   62   11
[2,]   11  122
#Matrices con diferente num. de filas y columnas
> ###-----
```

Problema 2.

Sean $A = \begin{pmatrix} 2 & 4 \\ -1 & 3 \end{pmatrix}$ y $B = \begin{pmatrix} 1 & 3 \\ 2 & -1 \end{pmatrix}$

- Encuentre AB y BA
- Encuentre $|A|$, $|B|$ y $|AB|$ (los determinantes)
- Obtenga $|BA|$ y compare con $|AB|$
- Obtenga la traza(AB) y compare con la traza (BA)
- Obtenga los eigenvalores de AB y de BA
- Obtenga el rango de A y de B
- Obtenga si existe la matriz inversa de A^{-1} y de B^{-1}

Rcode

```
> ###-----
> ##### Problema 2
> rm(list=ls())

> #Sean
> (A=matrix(c(2,-1,4,3),nrow = 2,ncol = 2))
      [,1] [,2]
[1,]    2    4
[2,]   -1    3
> (B=matrix(c(1,2,3,-1),nrow = 2,ncol = 2))
      [,1] [,2]
[1,]    1    3
[2,]    2   -1

> #   Encuentre AB y BA
> (AB=A%%B)
      [,1] [,2]
[1,]   10    2
[2,]    5   -6
> (BA=B%%A)
      [,1] [,2]
[1,]   -1   13
[2,]    5    5

> #   Encuentre |A|, |B| y |AB| (los determinantes)
> det(A)
[1] 10
> det(B)
[1] -7
> det(AB)
[1] -70

> #   Obtenga |BA| y compare con |AB|
> det(BA)
[1] -70
> det(AB)
[1] -70
```

```
> # Obtenga la traza(AB) y compare con la traza (BA)
> sum(diag(AB))
[1] 4
> sum(diag(BA))
[1] 4
# Es igual su traza
```

```
> # Obtenga los eigenvalores de AB y de BA
> eigen(AB)
eigen() decomposition
$`values`
[1] 10.602325 -6.602325

$`vectors`
      [,1] [,2]
[1,] 0.9575193 -0.1196004
[2,] 0.2883690 0.9928221

> eigen(BA)
eigen() decomposition
$`values`
[1] 10.602325 -6.602325

$`vectors`
      [,1] [,2]
[1,] -0.7460750 -0.9183529
[2,] -0.6658619 0.3957624
```

```
> # Obtenga el rango de A y de B
> qr(A)$rank
[1] 2
> qr(B)$rank
[1] 2
```

```
> # Obtenga si existe la matriz inversa de A(-1) y de B(-1)
> solve(A)
      [,1] [,2]
[1,] 0.3 -0.4
[2,] 0.1 0.2
> solve(B)
      [,1] [,2]
[1,] 0.1428571 0.4285714
[2,] 0.2857143 -0.1428571
```

Problema 3.

Sean $A = \begin{pmatrix} 5 & -1 & 3 \\ -1 & 1 & 2 \\ 3 & 2 & 7 \end{pmatrix}$, $B = \begin{pmatrix} 6 & -2 & 3 \\ 7 & 1 & 0 \\ 2 & -3 & 5 \end{pmatrix}$, $C = \begin{pmatrix} 2 & -3 \\ -1 & 4 \\ 3 & 1 \end{pmatrix}$,

$$x = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}, \quad y = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}, \quad z = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

Obtenga

- | | | |
|--------------|------------|--------------------|
| 1. Bx | 6. $x^t y$ | 11. zy^t |
| 2. $y^t B$ | 7. xx^t | 12. $\sqrt{y^t y}$ |
| 3. $x^t A x$ | 8. xy^t | 13. $B^t B$ |
| 4. $x^t C z$ | 9. $B^t B$ | |
| 5. $x^t x$ | 10. yz^t | |

Rcode

```
> ###-----
> ##### Problema 3
> rm(list=ls())
> #Sean
> (A=matrix(c(5,-1,3,-1,1,2,3,2,7),nrow = 3,ncol = 3))
      [,1] [,2] [,3]
[1,]    5   -1    3
[2,]   -1    1    2
[3,]    3    2    7

> (B=matrix(c(6,7,2,-2,1,-3,3,0,5),nrow = 3,ncol = 3))
      [,1] [,2] [,3]
[1,]    6   -2    3
[2,]    7    1    0
[3,]    2   -3    5

> (C=matrix(c(2,-1,3,-3,4,1),nrow = 3,ncol = 2))
      [,1] [,2]
[1,]    2   -3
[2,]   -1    4
[3,]    3    1

> (x=matrix(c(3,-1,2),nrow = 3,ncol = 1))
      [,1]
[1,]    3
[2,]   -1
[3,]    2

> (y=matrix(c(3,2,4),nrow = 3,ncol = 1))
      [,1]
[1,]    3
[2,]    2
[3,]    4
```

```
> (z=matrix(c(2,5),nrow = 2,ncol = 1))
      [,1]
[1,]    2
[2,]    5
```

```
> # Obtenga Bx
> (Bx=B%%x)
      [,1]
[1,]    26
[2,]    20
[3,]    19
```

```
> # y' B
> (yt=t(y))
      [,1] [,2] [,3]
[1,]    3    2    4

> (yt_B=t(y)%%B)
      [,1] [,2] [,3]
[1,]   40  -16   29
```

```
> # x' Ax
> (xt=t(x))
      [,1] [,2] [,3]
[1,]    3   -1    2

> (Ax=A%%x)
      [,1]
[1,]    22
[2,]     0
[3,]    21

> (xt_Ax=t(x)%%Ax)
      [,1]
[1,]   108
```

```
> # x' Cz
> (Cz=C%%z)
      [,1]
[1,]   -11
[2,]    18
[3,]    11

> (xt_Cz=xt%%Cz)
      [,1]
[1,]   -29
```

```
> # x' x
> xt%%x
      [,1]
[1,]    14
```

```
> # x' y
> xt%%y
      [,1]
[1,]    15
```

```
> #   xx'
> x%*%xt
      [,1] [,2] [,3]
[1,]    9   -3    6
[2,]   -3    1   -2
[3,]    6   -2    4
```

```
> #   xy'
> x%*%yt
      [,1] [,2] [,3]
[1,]    9    6   12
[2,]   -3   -2   -4
[3,]    6    4    8
```

```
> #   B'B
> (Bt=t(B))
      [,1] [,2] [,3]
[1,]    6    7    2
[2,]   -2    1   -3
[3,]    3    0    5
```

```
> Bt%*%B
      [,1] [,2] [,3]
[1,]   89  -11   28
[2,]  -11   14  -21
[3,]   28  -21   34
```

```
> #   yz'
> zt=t(z)
> y%*%zt
      [,1] [,2]
[1,]    6   15
[2,]    4   10
[3,]    8   20
```

```
> #   zy'
> z%*%yt
      [,1] [,2] [,3]
[1,]    6    4    8
[2,]   15   10   20
```

```
> #   raiz(y'y)
> sqrt(yt%*%y)
      [,1]
[1,] 5.385165
```

```
> #   B'B
> Bt%*%B
      [,1] [,2] [,3]
[1,]   89  -11   28
[2,]  -11   14  -21
[3,]   28  -21   34
```

Encuentre

- $x + y$
- $x - y$

Rcode

```
> #Encuentre  
> # x+y  
> x+y  
      [,1]  
[1,]    6  
[2,]    1  
[3,]    6
```

```
> # x-y  
> x-y  
      [,1]  
[1,]    0  
[2,]   -3  
[3,]   -2
```

Problema 4.

Escriba los sistemas de ecuaciones lineales en la forma $Ax = b$. Compare el rango de la matriz aumentada A^* con el rango del coeficiente matriz A para cada uno de los siguientes sistemas de ecuaciones, clasifique los sistemas de ecuaciones. Encontrar las soluciones donde existan

1. Sea el sistema

$$x_1 + 2x_2 + 3x_3 = 6$$

$$x_1 - x_2 = 2$$

$$x_1 - x_3 = -1$$

En la forma $Ax = b$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ -1 \end{pmatrix}$$

Rcode

```
> ###-----
> ### Problema 4
> rm(list=ls())

> # Inciso 1.
> #Matriz A
> (A=matrix(c(1,1,1,2,-1,0,3,0,-1),nrow = 3,ncol = 3))
  [,1] [,2] [,3]
[1,]  1  2  3
[2,]  1 -1  0
[3,]  1  0 -1
> (b=matrix(c(6,2,-1),nrow = 3,ncol = 1))
  [,1]
[1,]  6
[2,]  2
[3,] -1

> #Matriz A aumentada
> (A2=cbind(A,b))
  [,1] [,2] [,3] [,4]
[1,]  1  2  3  6
[2,]  1 -1  0  2
[3,]  1  0 -1 -1

> #Comparando rangos
> qr(A)$rank
[1] 3
> qr(A2)$rank
[1] 3
> #R. Rangos iguales - sistema compatible
```

```
> #Obtención de la solución
> (soluc=solve(A,b))
      [,1]
[1,]  1.1666667
[2,] -0.8333333
[3,]  2.1666667
> dimnames(soluc)<-list(c("x1", "x2","x3"), NULL)
> soluc
      [,1]
x1  1.1666667
x2 -0.8333333
x3  2.1666667
```

2. Sea el sistema

$$\begin{aligned}x_1 - x_2 + 2x_3 &= 2 \\x_1 - x_2 - x_3 &= -1 \\2x_1 - 2x_2 + x_3 &= 2\end{aligned}$$

En la forma $Ax = b$

$$\begin{pmatrix} 1 & -1 & 2 \\ 1 & -1 & -1 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

Rcode

```
> #-----
> #Inciso B
> rm(list=ls())
> (A=matrix(c(1,1,2,-1,-1,-2,2,-1,1),nrow = 3,ncol = 3))
      [,1] [,2] [,3]
[1,]    1    -1     2
[2,]    1    -1    -1
[3,]    2    -2     1
> (b=matrix(c(2,-1,2),nrow = 3,ncol = 1))
      [,1]
[1,]    2
[2,]   -1
[3,]    2

> #Matriz A aumentada
> (A2=cbind(A,b))
      [,1] [,2] [,3] [,4]
[1,]    1    -1     2     2
[2,]    1    -1    -1    -1
[3,]    2    -2     1     2

> #Comparando rangos
> qr(A)$rank
[1] 2
> qr(A2)$rank
[1] 3
> #R. Rangos diferentes - Sistema incompatible
```

3. Sea el sistema

$$x_1 + x_2 + x_3 + x_4 = 8$$

$$x_1 - x_2 - x_3 - x_4 = 6$$

$$3x_1 + x_2 + x_3 + x_4 = 22$$

En la forma $Ax = b$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & -1 \\ 3 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \\ 22 \end{pmatrix}$$

Rcode

```
> #Inciso C
> rm(list=ls())
> (A=matrix(c(1,1,3,1,-1,1,1,-1,1,1,-1,1),nrow = 3,ncol = 4))
      [,1] [,2] [,3] [,4]
[1,]    1    1    1    1
[2,]    1   -1   -1   -1
[3,]    3    1    1    1
> (b=matrix(c(8,6,22),nrow = 3,ncol = 1))
      [,1]
[1,]    8
[2,]    6
[3,]   22
```

```
> #Matriz A aumentada
> (A2=cbind(A,b))
      [,1] [,2] [,3] [,4] [,5]
[1,]    1    1    1    1    8
[2,]    1   -1   -1   -1    6
[3,]    3    1    1    1   22
```

```
> #Comparando rangos
> qr(A)$rank
[1] 2
> qr(A2)$rank
[1] 2
> #R. Rangos iguales - Sistema compatible
```

```
> #Obtención de la solución
> qr(t(A)%*(A))$rank
[1] 2
> qr(A)$rank
[1] 2
> library(MASS)
> ginv(t(A)%*(A))%*%t(A)%*%b
      [,1]
[1,] 7.0000000
[2,] 0.3333333
[3,] 0.3333333
[4,] 0.3333333
```